

Python AI Agent From Scratch · Logical Flow

· Project Overview

An AI research agent built with LangChain and Claude 3.5 Sonnet, providing web search, Wikipedia lookup, and file saving with structured Pydantic output.

- Complete Logical Flow

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Python AI Agent From Scratch
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User Research Question
    .
    .
    .
    .....
    . Claude 3.5 Sonnet Agent .
    .
    . Prompt Template: .
    . "Research the topic and provide .
    . a structured response" .
    . .
    . .
    . Agent Reasoning .
    . .
    . .
    . Tool Selection .
    .....
    .
    .
    .
    .....
    . DuckDuckGo . . Wikipedia . . Save .
    . Search . . Lookup . . to File .
    . . . . .
    . Web . . Summary . . .txt .
    . results . . articles . . output .
    .....
    .
    .
    .
    .....
    .
    .
    .
    .....
    . Pydantic Output Parser .
    .
    .
    . ResearchResponse: .
    . ... topic: str .
    . ... summary: str .
    . ... sources: List[str] .

```

```
. . ... tools_used: List[str] .  
. . ... confidence: float .  
. ....  
. .  
. .  
. Display Structured Research Report .  
. ....
```